

AY: 2025-2026
EXAM | AI-ECUE221
June 2026

M1-S2: Dept. of Electrical Engineering
Teacher: A. Mhamdi
Time Limit: 1½ h

This document contains 11 pages numbered from 1 to 11. Upon receiving it, verify completeness. The 3 tasks are independent and can be solved in any order you prefer. The following rules apply:

- ❶ A handwritten double-sided A4 sheet is permitted.
- ❷ Any electronic material, except basic calculator, is prohibited.
- ❸ Mysterious or unsupported answers will not receive full credit.
- ❹ Round results to the nearest thousandth (i.e., the third digit after the decimal point).
- ❺ Task N°3: Correct answers earn points as indicated. There is no negative scoring.



Task N°1

⌚ 35mn | (8 points)

Apply the DBSCAN algorithm to cluster the dataset shown in TAB. 1.

- ϵ (eps): 1.5 (maximum distance for neighborhood)
- MinPts: 3 (minimum number of points to form a dense region)

TAB. 1. Dataset with 10 points

Point ID	A	B	C	D	E	F	G	H	I	J
X	1	2	1	2	6	7	6	7	3	9
Y	1	1	2	2	6	6	7	7	5	2

- (a) (2 points) Calculate the distance matrix for all pairs of points¹.

Euclidean Distance Formula between two points $p_i(x_i; y_i)$ and $p_j(x_j; y_j)$ is:

$$\text{dist}(p_i, p_j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

Euclidean distances between all pairs of points:

¹Use Euclidean distance.

	A	B	C	D	E	F	G	H	I	J
A	0.0	1.0	1.0	1.414	7.071	7.81	7.81	8.485	4.472	8.062
B	1.0	0.0	1.414	1.0	6.403	7.071	7.211	7.81	4.123	7.071
C	1.0	1.414	0.0	1.0	6.403	7.211	7.071	7.81	3.606	8.0
D	1.414	1.0	1.0	0.0	5.657	6.403	6.403	7.071	3.162	7.0
E	7.071	6.403	6.403	5.657	0.0	1.0	1.0	1.414	3.162	5.0
F	7.81	7.071	7.211	6.403	1.0	0.0	1.414	1.0	4.123	4.472
G	7.81	7.211	7.071	6.403	1.0	1.414	0.0	1.0	3.606	5.831
H	8.485	7.81	7.81	7.071	1.414	1.0	1.0	0.0	4.472	5.385
I	4.472	4.123	3.606	3.162	3.162	4.123	3.606	4.472	0.0	6.708
J	8.062	7.071	8.0	7.0	5.0	4.472	5.831	5.385	6.708	0.0

The relevant Julia code is indicated in FIG. 1.

```
julia>
julia> d(α, β) = √((α[1]-β[1])^2+(α[2]-β[2])^2)
d (generic function with 1 method)

julia> data = [[1,1],[2,1],[1,2],[2,2],[6,6],[7,6],[6,7],[7,7],[3,5],[9,2]];

julia> using Printf

julia> for α in data
    for β in data
        @printf("%6.3f ", d(α, β))
    end
    println()
end
0.000 1.000 1.000 1.414 7.071 7.810 7.810 8.485 4.472 8.062
1.000 0.000 1.414 1.000 6.403 7.071 7.211 7.810 4.123 7.071
1.000 1.414 0.000 1.000 6.403 7.211 7.071 7.810 3.606 8.000
1.414 1.000 1.000 0.000 5.657 6.403 6.403 7.071 3.162 7.000
7.071 6.403 6.403 5.657 0.000 1.000 1.000 1.414 3.162 5.000
7.810 7.071 7.211 6.403 1.000 0.000 1.414 1.000 4.123 4.472
7.810 7.211 7.071 6.403 1.000 1.414 0.000 1.000 3.606 5.831
8.485 7.810 7.810 7.071 1.414 1.000 1.000 0.000 4.472 5.385
4.472 4.123 3.606 3.162 3.162 4.123 3.606 4.472 0.000 6.708
8.062 7.071 8.000 7.000 5.000 4.472 5.831 5.385 6.708 0.000

julia> █
```

FIG. 1. Distance Matrix (rounded to 3 decimal places)

(b) (1 point) For each point, find its ε -neighborhood.

For each point, find all points within distance 1.5:

Point	ε -Neighborhood
A	{A, B, C, D}
B	{A, B, C, D}
C	{A, B, C, D}
D	{A, B, C, D}
E	{E, F, G, H}
F	{E, F, G, H}
G	{E, F, G, H}
H	{E, F, G, H}
I	{I}
J	{J}

(c) (1 point) Classify each point as Core, Border, or Noise.

With $\text{MinPts} = 3$, classify each point:

Point	Neighborhood Size	Core?	Classification
A	4	Yes	Core
B	4	Yes	Core
C	4	Yes	Core
D	4	Yes	Core
E	4	Yes	Core
F	4	Yes	Core
G	4	Yes	Core
H	4	Yes	Core
I	1	No	Noise
J	1	No	Noise

(d) (2 points) Apply the DBSCAN algorithm step by step to identify the final clusters and noise points.

- Start with point A:**
 - A is core $\rightarrow$ Create Cluster 1
 - Add all density-reachable points: {A, B, C, D}
 - Mark A, B, C, D as visited
- Continue with unvisited points:**

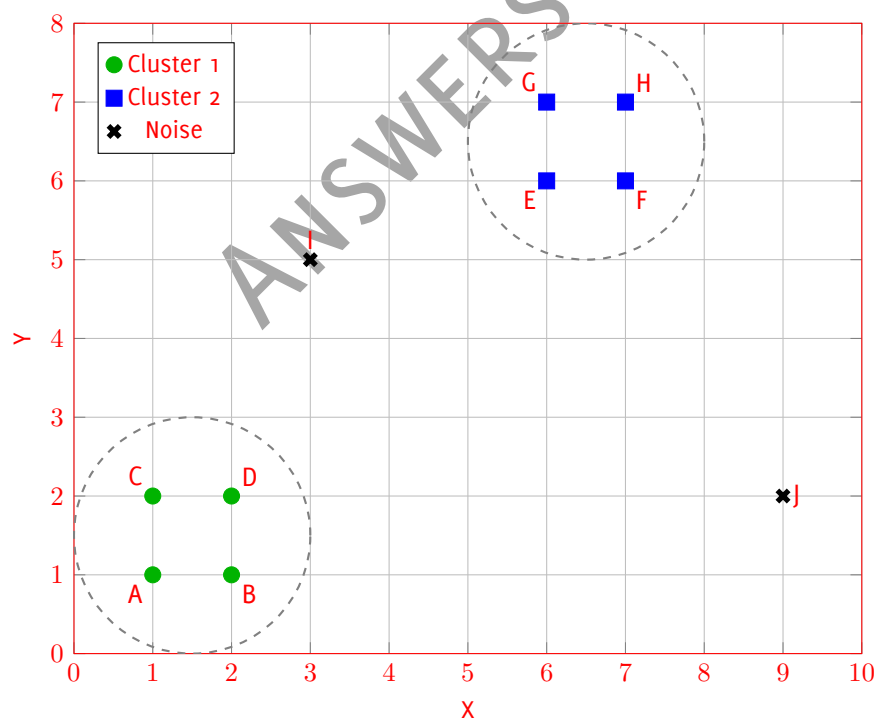
- Next unvisited: E
- E is core $\rightarrow$ Create Cluster 2
- Add all density-reachable points: {E, F, G, H}
- Mark E, F, G, H as visited

3. Continue with remaining points:

- Point I: Not core, not in any core's neighborhood $\rightarrow$ Noise
- Point J: Not core, not in any core's neighborhood $\rightarrow$ Noise

Cluster	Points
Cluster 1	{A, B, C, D}
Cluster 2	{E, F, G, H}
Noise	{I, J}

(e) (2 points) Draw the clustering result.



The DBSCAN algorithm with parameters $\varepsilon = 1.5$ and $\text{MinPts} = 3$ successfully:

1. Identified two distinct clusters: one in the bottom-left region (points A, B, C, D) and one in the top-right region (points E, F, G, H);
2. Classified two isolated points (I and J) as noise.

Build a decision tree classifier using the Gini index as the split criterion for the data in TAB. 2. The goal is to predict the class (A or B) based on the given features.

TAB. 2. Training Dataset (12 samples)

Age	Income (k\$)	Student	Class
25	35.000	Yes	A
35	50.000	No	B
22	25.000	Yes	A
45	80.000	No	B
28	40.000	Yes	A
50	90.000	No	B
30	45.000	No	B
24	30.000	Yes	A
40	70.000	No	B
26	35.000	Yes	A
55	95.000	No	B
23	28.000	Yes	A

(a) (1 point) Calculate the Gini index for the root node.

The Gini index for a dataset D is defined as:

$$G_{\text{Root}} = 1 - \sum_{i=1}^c p_i^2$$

where p_i is the proportion of samples belonging to class i , and c is the number of classes.

Information Gain: The information gain for a split is calculated as:

$$G(D, A) = G(D) - \sum_{v \in \text{Values}(A)} \frac{|D_v|}{|D|} \times G(D_v)$$

where D_v is the subset of D for which attribute A has value v .

The class distribution in the entire dataset is:

- Class A: 6 samples
- Class B: 6 samples
- Total: 12 samples

Calculate the proportions:

$$p_A = \frac{6}{12} = 0.5$$
$$p_B = \frac{6}{12} = 0.5$$

Calculate the Gini index:

$$\begin{aligned} \text{Gini}_{\text{Root}} &= 1 - (p_A^2 + p_B^2) \\ &= 1 - (0.5^2 + 0.5^2) \\ &= 1 - (0.25 + 0.25) \\ &= 1 - 0.5 \\ &= 0.5 \end{aligned}$$

(b) (3 points) Find the best split by evaluating all possible splits for each feature node.

When splitting on continuous features at any node:

- Calculate the median of the feature values in the current node
- Create a binary split where:

Left child: Instances with feature value $\leq$ median;

Right child: Instances with feature value $>$ median.

Age Feature Splits

Split: Age $\leq$ 29 (median age)

- Left (Age $\leq$ 29): {22, 23, 24, 25, 26, 28} $\rightarrow$ 6 samples, all Class A
- Right (Age $>$ 29): {30, 35, 40, 45, 50, 55} $\rightarrow$ 6 samples, all Class B

Gini calculations:

$$\text{Gini}(\text{Left}) = 1 - \left(\left(\frac{6}{6} \right)^2 + \left(\frac{0}{6} \right)^2 \right) = 0$$

$$\text{Gini}(\text{Right}) = 1 - \left(\left(\frac{0}{6} \right)^2 + \left(\frac{6}{6} \right)^2 \right) = 0$$

Weighted Gini:

$$\text{Weighted Gini} = \frac{6}{12} \times 0 + \frac{6}{12} \times 0 = 0$$

Information Gain:

$$\text{Gain} = 0.5 - 0 = \boxed{0.5}$$

Income Feature Splits

Split: $\text{Income} \leq \text{k\$42.500}$ (medial income)

- Left: 6 samples (6 Class A, 0 Class B)
- Right: 6 samples (0 Class A, 6 Class B)

Gini calculations:

$$\begin{aligned}\text{Gini}(\text{Left}) &= 1 - (1^2 + 0^2) \\ &= 0\end{aligned}$$

$$\text{Gini}(\text{Right}) = 1 - (0^2 + 1^2) = 0$$

Weighted Gini:

$$\begin{aligned}\text{Weighted Gini} &= \frac{6}{12} \times 0 + \frac{6}{12} \times 0 \\ &= 0\end{aligned}$$

Information Gain:

This gives the same result as the Age split:

$$\text{Gain} = 0.5 - 0 = 0.5$$

Student Feature Split

Split: $\text{Student} = \text{Yes}$

- Left (Student = Yes): 6 samples, all Class A
- Right (Student = No): 6 samples, all Class B

Information Gain:

$$\text{Gain} = 0.5 - 0 = \boxed{0.5}$$

(c) (1 point) Calculate the information gain for the best split.

Comparing all splits:

- Age ≤ 29 : Gain = 0.5
- Income $\leq \$42,500$: Gain = 0.5
- Student = Yes: Gain = 0.5

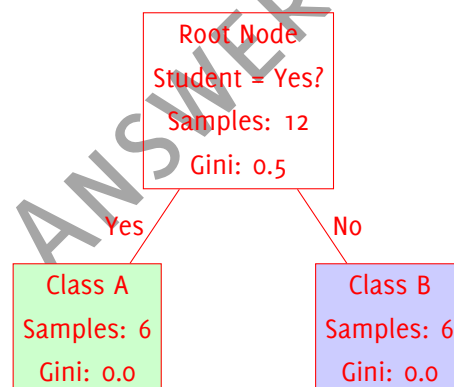
All features achieve the maximum gain of 0.5. We choose **Student = Yes** for simplicity.

(d) (1 point) Build and draw the complete decision tree.

Since both resulting nodes are pure (Gini = 0), no further splitting is required.

Final Tree:

- Root: Student = Yes?
- Left branch (Yes): Predict Class A (6/6 samples)
- Right branch (No): Predict Class B (6/6 samples)



(e) (1 point) Using the decision tree you built in the previous steps, classify the following new unseen data points:

TAB. 3. New Data Points for Classification

New Point	Age	Income (k\$)	Student	Predicted Class
P1	27	38.000	Yes	?
P2	42	65.000	No	?
P3	29	55.000	Yes	?
P4	35	45.000	No	?

Since our decision tree splits solely on the “Student” feature, the classification is straightforward:

- All students (Student = Yes) are predicted as Class A
- All non-students (Student = No) are predicted as Class B
- The Age and Income features are not used in this particular tree, demonstrating that the Student feature alone provides perfect separation for the training data

Alternative Analysis: If we had chosen the Age split instead:

- **Rule 1:** If Age $\leq 29 \rightarrow$ Predict Class A
- **Rule 2:** If Age $> 29 \rightarrow$ Predict Class B

This would yield identical performance, demonstrating that multiple optimal splits can exist in decision tree construction.

ANSWERS

AY: 2025-2026

Full Name:

M1-S2: Dept. of Electrical Engineering

ID:

EXAM | AI-ECUE221

Class: RAIA1

June 2026

Room:

Teacher: A. Mhamdi

Time Limit: 1½ h



ANSWER SHEET

Task N°3

⌚ 25mn | (5 points)

- (a) (½ point) What distinguishes supervised from unsupervised learning?
- ☐ Supervised uses unlabeled data, unsupervised uses labeled data
 - ☐ Supervised is only for regression, unsupervised for classification
 - ☐ Unsupervised always has better performance than supervised
 - ✓ Supervised requires target variables, unsupervised discovers patterns
- (b) (½ point) Which of the following is NOT a typical application of unsupervised learning?
- ✓ Image classification
 - ☐ Dimensionality reduction
 - ☐ Anomaly detection
 - ☐ Customer segmentation
- (c) (½ point) Consider a hard-margin SVM trained on a linearly separable dataset. Let the optimal solution to the dual problem be given by Lagrange multipliers α_k . Which of the following statements is correct?
- ☐ All training points have $\alpha_k > 0$
 - ☐ Only misclassified points have $\alpha_k > 0$
 - ✓ Only support vectors have $\alpha_k > 0$
 - ☐ Points far from the margin have the largest α_k
- (d) (½ point) Suppose you are using an SVM with a kernel $\mathcal{K}(\mathbf{x}_k, \mathbf{x}_l)$. You never explicitly compute the feature mapping $\Phi(\mathbf{x})$. Which statement is true?
- ☐ The dimensionality of $\Phi(\mathbf{x})$ must be finite
 - ☐ The weight vector $\mathbf{w}$ is explicitly computed in input space
 - ☐ The decision boundary is always linear in input space
 - ✓ The classifier depends only on dot products between data points



- (e) ($\frac{1}{2}$ point) Which of the following is a key characteristic of K-Means clustering?
- ☒ It requires the number of clusters to be specified in advance
 - ☐ It automatically determines the number of clusters based on the data
 - ☐ It can handle non-spherical cluster shapes well
 - ☐ It is a density-based clustering method
- (f) ($\frac{1}{2}$ point) What is the primary advantage of DBSCAN over K-Means?
- ☒ It can find arbitrarily shaped clusters and identify noise
 - ☐ It is faster for large datasets
 - ☐ It requires no parameters
 - ☐ It works better with high-dimensional data
- (g) ($\frac{1}{2}$ point) In PCA, what is the purpose of the eigenvectors of the covariance matrix?
- ☐ They indicate the mean of the data
 - ☐ They represent the class labels in the transformed space
 - ☒ They represent the directions of maximum variance in the data
 - ☐ They are the residuals of the projection
- (h) ($\frac{1}{2}$ point) Which of the following statements about PCA is FALSE?
- ☐ The first principal component captures the direction of the maximum variance in the data
 - ☐ PCA is sensitive to the scaling of the features
 - ☐ PCA is an unsupervised dimensionality reduction technique
 - ☒ The principal components are not orthogonal to each other
- (i) ($\frac{1}{2}$ point) Julia's advantage for ML workflows includes:
- ☐ Exclusive GPU acceleration support
 - ☒ Just-in-time (JIT) compilation and multiple dispatch
 - ☐ Python syntax compatibility
 - ☐ Built-in deep learning frameworks
- (j) ($\frac{1}{2}$ point) Key advantage of SGD over batch gradient descent:
- ☐ Guaranteed global optimum
 - ☐ Higher precision in large datasets
 - ☒ Lower computational cost per iteration
 - ☐ Eliminates all local minima